Study of Radical Views on Social Media: Classification and Group Dynamics Analysis

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***Abstract* — Social media platforms have changed the way extremist groups recruit, influence and potentially radicalize users online. Radicalization has drawn the attention of many researchers worldwide with its increasing presence on social media. Therefore, the ability to identify and classify radical views online and their potential role in radicalizing individuals is of critical importance. In this work, we address this challenge by combining a number of machine learning and natural language processing techniques such as sentiment analysis, named entity recognition, clustering, opinion leader identification, and social circle analysis to gain insights about radical views related to extreme nationalism, and understand the social dynamics underpinning extremists' related interactions. The proposed approach was implemented and tested on more than 259,000 tweets collected over a five-year span. The different analysis steps were presented and the results obtained were analyzed.**

Keywords— Radical views, Social media analytics, nationalism, group dynamics, classification and clustering

I. INTRODUCTION

Radicalization nowadays is mostly mentioned with reference to the radicalization of non-state actors challenging the state (McCauley & Moskalenko 2008). Since 9/11 there has been an explosion of the literature on radicalization in relation to radical Islam. The latest noticeable trend as noted by the Global Terrorism Index [1] is the rise of far-right terrorism and it seems we can expect as well more research on this particular form of political violence.

Radicalization is a multidimensional concept hence the plurality of definitions available in the literature. Multicausality of radicalization makes its study particularly complex [2]. Politicization related to the security agenda in the West and the relativism of the term -depending on the context and circumstances in which it is considered- add confusion [2]. The usefulness of the concept itself is also met with skepticism by some scholars especially because it overlooks the role of ideas and action, disproportionately relying on radical ideology as the main element pushing an individual to political violence [3]. Sedgwick [2] advises researchers "to abandon the idea that ''radical'' or ''radicalization'' are absolute concepts, to recognize the essentially relative nature of the term ''radical,'' and to be careful always to specify both the continuum being referred to and the location of what is seen as ''moderate'' on that "continuum". Ozer and Bertelsen [4] also rightly point at the difficulty of studying radicalization because it differs depending on the context. They identify five types of extremism: nationalistic or separatist, extreme right-wing, extreme left-wing, specific single issue and religiously motivated but all of them are geared towards change into society and belief in violence as a means to this end.

McCauley and Moskalenko [3] provide an excellent review of the different definitions of radicalization which they themselves view as a "change in beliefs, feelings, and behaviors in directions that increasingly justify intergroup violence and demand sacrifice in defense of the ingroup" [3]. Perceptions and understandings of radicalization as a process are prevalent.

McCauley and Moskalenko [3] first refer to Horgan who emphasizes the gradual progression to terrorism from legal activism to violent action, the sense of dissatisfaction of an individual making them more vulnerable to influence and finally, the community support for violent action. McCauley and Moskalenko [3] also refer to Wiktorowicz four-stage model of radicalization which includes: 1) a cognitive opening to new ideas and people following experiences of grievance; 2) a personal connection with activists; 3) the acceptance of a new legitimate authority; and 4) a total belief in violent action.

In their literature review, Doosje et al. [5] identify factors of radicalization at different levels (micro, meso, macro) active in the three phase of radicalizations: 1) a "sensitivity phase" determining sensitivity to being influenced; 2) a "group-membership phase" where an individual joins a radical group; 3) an action-phase corresponding to the active turn to violence.

The role of social media is now crucial with regard to radicalization: it is relatively easy for recruiters to navigate social media in order to identify potential candidates and concentrate their efforts on them. In the case of Islamic radical groups, while in the past, religious leaders would typically motivate terrorist fighters, fighters now prefer to rally around a charismatic leader, usually a former jihadist from the Afghan front against the Soviet Union [6]. "New spiritual authorities" operate on the Internet and act as inspirational figures, representing the emergence of a new style of virtual preacher without borders. These disseminators are not even necessarily in the field, but they act on their own initiative for the cause of Daesh [6].

Twitter is precisely a major tool for disseminators [7]. Facebook accounts give the possibility to share all the aspects of the jihadi life in Syria. The radical discourse is thus

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appealing to emotions and can reach greater numbers through the Internet. Even humor and the codes of geek and pop culture are used in the process, in order to normalize the status of these terrorist fighters and thus attract even more potential candidates as the tech-savvy generation is the main target of Daesh [8]. Dominant tools explored in the literature are compiled in Agarwal [9] and Correa & Sureka [10]:

Numerous studies, especially in the wake of ISIS revolutionary use of the Internet, have explored the role of social media in encouraging radicalization [11]. Lara-Cabrera et al. adopt several indicators to measure radicalization online in the context of Muslims in the West and based on pro-ISIS Twitter accounts with a timeline starting in November 2015 (Paris attacks) [11]. Cross and Logan (2013) coded the websites of 15 extremist groups categorized as activists, radicals, or terrorists based on the extremity of opinion (quoted in McCauley and Moskalenko' two-pyramid model 2017, 212). The RiskTrack project 1 funded by the European Commission objective is to "identify and tackle those factors or indicators that raise a red flag about which individuals or communities are being radicalised online and recruited in the view to commit violent acts of terrorism". In the framework of this project, Torregrosay et al. (2017) raise the question of whether radicalization can be measured analyzing users' information on the Internet and present a case-study using Facebook to see whether it is possible to identify indicators of jihadist radicalization. The analysis was conducted manually by using a psychological risk tool. In the framework of this project, the research team developed five indicators "to measure the online radicalisation assessment of a given individual using public data from his social networks" [11].

The analysis relies on an extraction model doing quantitative analysis about written text in social networks such as tweets. There are significant correlations between the expression of "positive ideas about Jihadism, the perception of discrimination, the use of swear words, and the use of negative words" [11].

Several machine learning techniques may be used in these studies for detection of radicalization online. A report by the Practices Project lists several of these studies designed to detect radicalization online (pages 10-13), including Link Based Bootstrapping (LBB) or the use of Bayesian classifiers. Correa and Sureka provide a review of the different techniques, analyzing 40 papers in "Solutions to detect and analyze online radicalization: a survey" (2013). Correa & Sureka [10] and Agarwal [9] provide useful indicators on tools used for such research in their respective literature reviews.

Aiming at contributing of the area of Extremism analysis and classification, we propose the use of machine learning and natural language processing (NLP) techniques to analyze social media content, in order to identify extreme views and radical beliefs. Our proposed framework leverages sentiment analysis, named entity recognition, clustering, opinion leader identification, and social circle analysis to gain insights about extreme speech and the social dynamics of individuals with radical views. Our proposed approach was implemented and tested using 259,000 tweets, posted over 74 months, in relation to extreme nationalism. The results presented are very promising and open the door to sophisticated social media analytics in relation to extreme speech and its use for the radicalization of individuals.

The rest of the paper is organized as follows: Section II provides some background information and discusses related work on methods that can be used for the classification of online radicalization. Section III details our proposed approach and analyses the results obtained. We end with our conclusions in Section IV.

II. RELATED WORK

The analysis and study of online radicalization is a very active area of research. In the sub-sections below, we briefly overview the most relevant research focusing on classification based on behavior, sentiment analysis, link analysis, identification of opinion leaders, and temporal behavior analysis.

A. Online Radicalization Classification based on Behavior

Raul Lara-Cabrera et al. [11] introduces a model with the main goal of being able to identify and validate individual users on social media platforms with a high risk of being radicalized [11]. They adopt a machine-learning method that utilizes 5 different behavioral indicators set by expert psychologists to identify each individual user. In the article, the authors focus only on English language tweets, however; the data extracted identified different languages (Arabic, French, etc.…) but were not taken into consideration which can be identified as a limitation.

B. Online Radicalization Classification based on Sentiment Analysis

Hsinchun Chen [12] focuses on presenting an automated approach for sentiment and affect analysis to identify radicalization in Dark Web Forums. Similar to [11][9] the following approach incorporates machine learning techniques and analyzing the context of the text in the messages to identify the sentiment polarity and affect intensity In comparison to 4 main affects: violence, anger, hate and racism [8]. Furthermore, Ryan Scrivens et al. [13] concentrate on identifying the "most" radical users present on the Dark Web from 4 different Islamic forums with the help of sentiment analysis in addition to a novel algorithm known as sentiment-based Identification of radical authors (SIRA) [13]. By collecting the necessary data from the different forums and applying Parts of speech (POS) analysis, the output results in a keyword list for the sentiment analysis to help in identifying the polarity value for each individual post. Whilst the overall output of the experiment was able to identify the same radical user across two separate forums. The following experiment was based on a limited set of user accounts, which can make it difficult to generalize the results in a more complex scenario.

C. Online Radicalization Classification based on Link Analysis

Yilu Zhou et al. [14] conducted a research focusing on understanding how US domestic extremist groups are using

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the Internet specifically online web pages and how to automate analyzing and visualizing individual content, by collecting data from approximately 97 URLs. Link analysis is used to calculate any similarity between different website pairs, in addition to using multidimensional scaling (MDS) algorithm, and finally content analysis to capture content on the websites and obtain the attribute with the highest frequency. Whilst the link-analysis did identify inter-connections amongst some communities and the highest attribute identified was terrorism and hate ideology, however; the following research only focused on a specific region only and extracted information from a small sample of URLs. Similarly, Hsinchun Chen et al. [12] developed a semi-automated methodology in identifying and collecting Dark web information from different Jihad websites incorporating information collection, analysis, human domain knowledge, and visualization techniques [12]. Such that affiliations, types of terrorist use and attributes were identified, recorded, and visualized using an MDS and snowflake diagram. The output of the diagrams results that terrorists rely heavily on sharing their ideology on the web and different clusters are linked to one another.

D. Identification of Opinion Leaders

Following the new virtual style of rallying around a charismatic leader. Abolfazl Aleahmad et al. [15] proposed a new algorithm (OLFinder) that identifies any important topic discussed in a given domain, and calculates the probability of each user being an opinion leader who may play an important role in influencing the minds of their followers [15]. Similarly, Fatia Kusuma Dewi et al. [16] proposes another method of identifying opinion leaders by conducting an edge weighting experiment, centrality weighting experiment and which results from each experiment would give an accurate classification of an opinion leader.

E. Temporal Behavior analysis

Ferrara et al. [17] came up with a machine learning framework that considers data collected from Twitter to be able to identify extremist users, to predict the spread of extremist content, and to predict interactions between extremist users and other users over time. Similarly, Pooja Wadhwa et al. [18] focuses on investigating the dynamic behavior of radical groups present in social networks mainly twitter by implementing data mining techniques such as textual analysis focusing on the context of the text in the messages in addition to the application of techniques used in social network analysis to discover the evolution and tracking of communities over time [9]. However, one suggested limitation to the approach is the shortness duration of testing the model (set to a single day).

III. Extreme Views Classification Methodology and Results

In order to build a framework for extreme views classification, we follow six main steps, as depicted in figure 1.

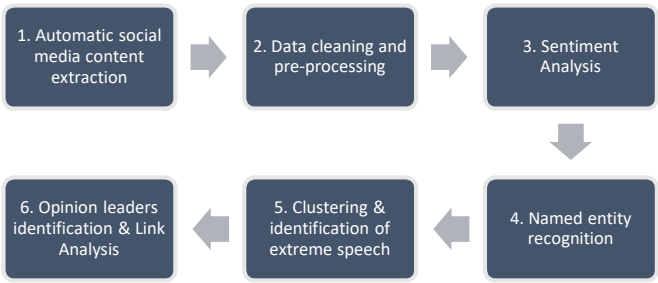

Fig. 1. Extreme Views Classification Framework

A. Dataset Collection

To create a dataset for our study, twitter API search was used. We selected specific hashtags related to extremism, which were identified by examining anchor websites. The data collected includes tweets posted over the span of five years (from 2016 to 2021). In addition to the tweets text, other fields of interest were collected, such as: Twitter user's information, user's mentions, user's location, user's number of followers, and user's number of friends. We will now detail the stages of data collection employed: The first stage was a bulk data extraction based on a set of hashtags. The list of the hashtags that were used to extract the data are: #nomigrants, #protectourborders, #MAGA, #whitegenocide, #chinesevirus, #nigger, #whitelivesmatter, #whiteoutwednesday, #chinavirus, #stopislam, #islamterrorist, #buildthewall, #nigger, and #notracist. The total dataset extracted consists of the posts of ***10 opinion leaders and their followers*** including ***259,904 total tweets***, posted from ***21-02-2016 until 01-05-2021***. The characteristics of our analysis dataset are illustrated in Table 1.

Table 1. Characteristics of our dataset

Number of Opinion Leaders	10
Total Number of tweets	259,904
Total number of followers	612,534,930
Total number of likes	381,369
Total Number of retweets	124,128,861
Total number of replies	39,291
Start date	21/02/2016
End Date	01/05/2021

B. Data cleaning and pre-processing

Tweets captured are filtered based on language to ensure that only English language tweet texts are kept while other languages are removed as they will be out of scope for this research. Additionally, any hyperlinks present, any emojis present, any retweets present will be removed to reduce processing overhead and clean the dataset. Following the filtering process, each tweet text will be tokenized such that a token will correspond to each word. Following tokenization is further processing for removal of stop words present in these tokens such as "a, the, is, of, are, etc.…". Finally, the final preprocessing step following the removal of stop words is lemmatization. Lemmatization is useful as it considers the context of the words and converts the word to its original meaningful base form, for example 'terrorist' can be identified from the root word 'terrorism'. Table 2 illustrates

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the data cleaning and pre-processing step, when applied to one of the collected tweets.

Table 2. Illustration of data cleaning and pre-processing

Tweet Text	Clean Tweet No Emojis	Tweet Tokenized	Tweet No Stop Words	Tweet Lemmatized
Itâ€™s sad the only place you find white people in the black community protesting is at the plan parenthood! Imaginâ€¦ https://t.co/dFhwG21h5i	Its sad the only place you find white people in the black community protesting is at the plan parenthood! Imagin	['its', 'sad', 'the', 'only', 'place', 'you', 'find', 'white', 'people', 'in', 'the', 'black', 'community', 'protesting', 'is', 'at', 'the', 'plan', 'parenthood', 'imagin', '']	['sad', 'place', 'find', 'white', 'people', 'black', 'community', 'protesting', 'plan', 'parenthood', 'imagin', '']	['sad', 'place', 'find', 'white', 'people', 'black', 'community', 'protesting', 'plan', 'parenthood', 'imagin', '']

C. Sentiment Analysis

Sentiment analysis is leveraged in this research to identify the polarity of each tweet text. The polarity focuses on emotions present by the user such as joy, sad, happiness, and anger [18]. The first step is to ensure the tweet text, which is the field that is going to be analyzed for sentiment, is free of all unnecessary characters and URLs, is tokenized, and is free of stop words. The next stage is stemming and lemmatization prior to evaluating the tweet text for polarity. Sentiment polarity ranges between -1,0,1. If the polarity is equal to 0 then the tweet text is neutral, if the polarity is greater than zero then the tweet is positive, else the tweet is negative. Table 3 illustrates the results of the sentiment analysis process for the example tweet pre-processed in the first step.

Table 3. Illustration of tweet's sentiment analysis

Tweet Text	Tweet Lemmatized	Polarity	Score
Itâ€™s sad the only place you find white people in the black community protesting is at the plan parenthood! Imaginâ€¦ https://t.co/dFhwG21h5i	['sad', 'place', 'find', 'white', 'people', 'black', 'community', 'protesting', 'plan', 'parenthood', 'imagin', '']	-0.22222222 22222222	Negative

D. Named Entity Recognition (NER)

Similarly, Name entity recognition, a common text extraction analysis that helps in identifying and classifying named entities present in the unstructured tweet text into pre-defined labels such as: organization, person name, location, date, etc..... The aim of the NER analysis is to help in identifying and classifying the level of extremism present in the tweet text for each opinion leader present. To achieve such a goal, further refinement was applied to the common NER algorithm by adding additional named entity labels with their associated named entities as illustrated in Table 4. To identify the level of extremism available a counter is set to iterate within the presence of the name entity labels respectively, and a percentage of extremist tweets per month for every opinion leader will be identified accordingly. Based on the calculated percentage. If the value is 20% or lower, the individual is not considered as extremist, if the value is between the range of 20%-50% then the individual is considered to be a moderate extremist, and if the value is 50% or greater than the individual is considered to be highly extremist.

A snippet of the NER algorithm's output is shown in figure 2. As shown, the enhanced NER algorithm has successfully identified specific terms associated with extremism such as 'whitelivesmatter' and 'whitegenocide' as racism and whitesupremacism.

Table 4. Extremism related NER labels introduced

Name Entity Labels	**Name Entities**
White Supremacy	"whitesupremacy", "whiteprivilege","whiteterrorism","whitegenocide","whitepower","whitediscrimination","gop", "trumpism", "zionists", "anti-semitism"
Islamophobia	"jihad","stopislam","islamterrorist","isis","radicalIslam","islamophobia","extremist", "mujahedeen","discrimination", "anti-muslimism"
Xenophobia	"chinesevirus","chinavirus","asian","chinese"
Anti-Immigration	"refugeesnotwelcome","protectourborders","buildthewall","abolishICE","nobannowall", "ban","radical","refugee","criminals","rapists","immigrants"
Racism	"nigga", "nigger", "whitelivesmatter", "whiteoutwednesday", "notracist"

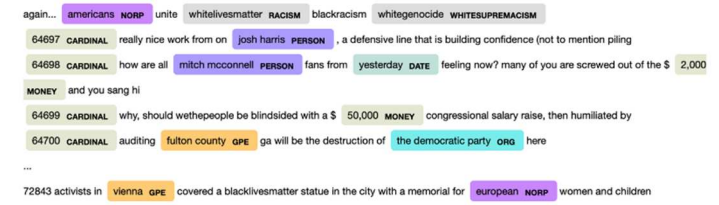

Fig. 2. Illustration of NER result

E. Clustering and Identification of Extreme Speech

One of the main objectives of this work is to classify social media posts into extreme or non-extreme speech, using machine learning. Relying on supervised learning would require manual labelling of our full dataset consisting of hundreds of thousands of tweets, which is a lengthy and tedious process. Instead, we chose to rely on an unsupervised clustering approach which can automatically uncover meaningful groupings in our data, without the need for manual labelling. Once clustered, we then apply NER on the clusters' related content to calculate the percentage of extreme speech in them, thus identifying relevant clusters that reflect extreme views. Those clusters can then be used to undergo further analysis.

In this work, we relied on the K-means clustering technique that groups tweets together based on their similarity, to classify tweets as extreme or not. K-means clustering is a popular unsupervised machine learning technique used to group categories together based on the specific identified number of centroids. Mini Batch K-means clustering was leveraged in this study to for an improved clustering performance. Prior to clustering, a numerical statistical technique TF-IDF (term frequency- inverse document frequency) was computed on the tweet texts to help in extracting important/unique words that will help in the clustering process. Following the successful transformation of the tweet text into a TF-IDF score, it is essential to specify the number of clusters needed, and so a common approach is plotting the SSE (sum of squared estimate of errors) for a range of cluster sizes and eventually choose the "elbow" where SSE begins to level off. As shown in figure 3, the elbow

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method suggests eight as optimal number of clusters for our dataset.

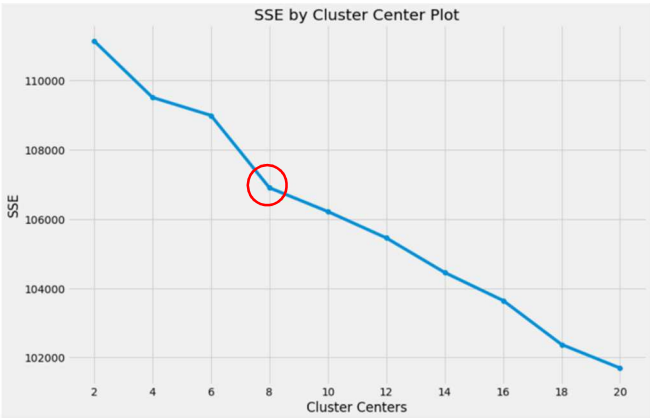
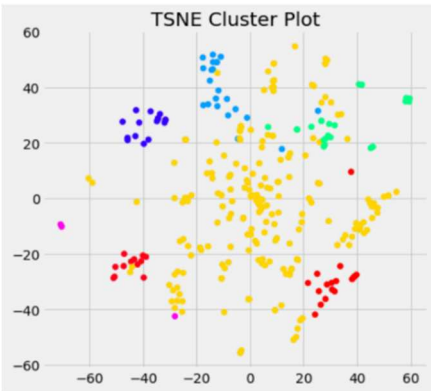
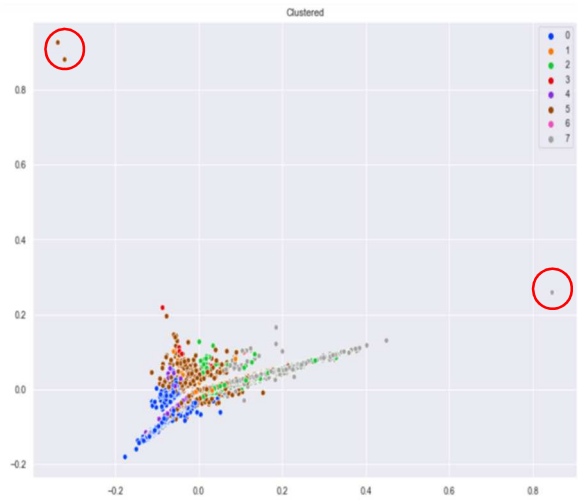


Fig. 3. SSE plot for our dataset

To visualize the obtained data clusters and reduce its dimensionality, two techniques were used: 1) PCA (principal component analysis) which is a dimensionality reduction method able to capture the global structure of the data; and 2) TSNE (t-distributed Stochastic Neighbor Embedding) which helps in visualizing high dimensional data and enables capturing of the relations between neighbors. Figure 4 depicts the clusters' visualization results. As observed, some clusters (cluster 0,2,5, and 7) overlap with each other and that there are some outliers present in the dataset.



(a)



(b)

Fig. 4. Visualization of data clustering results: (a) TSNE clusters' plot; (b) PCA clusters' plot

Cycling through the clusters identified, our system then prints out the top keywords present in each cluster based on the TF-IDF score to identify any possible trends visible and the most relevant clusters related to extremism. The keywords associated with the obtained clusters are shown in table 5.

Table 5. Keywords associated with identified clusters

Clusters	**Top Keywords**
Cluster 0	arm,build,win,trump2020,maga,protectourborders,trump,border,wall,buildthewall
Cluster 1	like,protectourborders,people,trump,amp,china,whitelivesmatter,nigger,chinavirus,maga
Cluster 2	retweet,seen,buy,lady,young,monday,afternoon,bread,southafricancoro,hillbrow
Cluster 3	covering,horrible,tag,fan,stans,trending,http,whitelivesmatter,kpop,whiteoutwednesday
Cluster 4	whitelivesmatter,non,race,genocide,country,diversity,people,anti,whitegenocide,white
Cluster 5	thisiswhyiresist,talk,especially,people,wealthy,nah,said,thing,white,whiteprivilege
Cluster 6	china,identified,news,check,coronavirus,rt,share,covid 19,chinavirus,covid19
Cluster 7	growing,chinese,africa,south,covid19,southafricancoronavirus,virus,true,china,chinesevirus

Analyzing the obtained keywords, we observe the following:

- Five main themes emerged from the data clusters. A shown in table 6, the anti-immigration theme was observed in clusters 0 and 1, the white supremacy theme was observed in clusters 1 and 4, and anti-Asian racism theme was observed in clusters 1, 6, 7. Other themes such as Antifa and South Africa were observed in clusters 3&5, and 2&7 respectively.
- Cluster 1 is interesting and very diverse. It provides evidence that right-wing extremism does not have a clear ideology and is a mixed galaxy of conspiracy theories, religious elements, xenophobia, racism, and white supremacy.
- Cluster 2 exclusively concerns South Africa, while cluster 3 seems to refer to a K-Pop fans operation to sabotage the hashtag #whitelivematters. Those two clusters would not be reliable to analyze extremism.
- Cluster 5 is from antifa (white privilege denunciation, whyIresist hashtag), and thus should not be put at the same level as any potential radicalism of antiracist movements and r acist movements.
- For anti-Asian racism, cluster 6 is more interesting than cluster 7 which is less focused on COVID as a "China virus" and contains mentions of South Africa.

Based on the above analysis, the most relevant clusters to consider would be ***clusters 0, 4, 6 and cluster 1*** which is transversal as it encompasses various themes.

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Table 6. Dominant themes in identified clusters

Anti-immigration	White supremacy	Anti-Asian racism	*Antifa*	*South Africa*
Cluster 0 Cluster 1	Cluster 4 Cluster 1	Cluster 6 Cluster 7 Cluster 1	*Cluster 5* *Cluster 3*	*Cluster 2* *Cluster 7*

F. Opinion Leaders Identification and Link Analysis

Building on the results of the clustering phase, and to obtain more insights about extremism and extremist social dynamics, we focused on identifying opinion leaders in the most interesting clusters, and conducted link analysis of their social networks on Twitter.

Opinion leaders are influential figures characterized by their popularity and competency in a certain domain. In this work, we leveraged the eigenvector centrality algorithm to identify the opinion leaders in the clusters of interest. Afterwards, we conducted link analysis of those opinion leaders and their followers, and plotted their social circle graphs on Twitter.

The social circle graph is obtained by iterating through each opinion leader and a portion of their respective followers and generating an edge list that contains each user and each follower paired one to one. Initially, this requires the extraction of a random sample of the followers and obtaining their screen name. Then, a loop goes through each user and their selected followers, placing them within a single list that is a one-to-one list of the users and their followers. The networkx library is then used to turn this into a graph of a network that is filtered to include nodes with a minimum of two or three edges (depending on the opinion leader), as in the follower's screen name must have been listed at least twice in the list. Figure 5 depicts the social circle graphs of five of the identified opinion leaders.

Link analysis allows us to determine social communities, visualize the degree of connectivity and the location of both opinion leaders and opinion-seekers (followers). The location of nodes and number of edges can allow us to make inferences about centrality (and thus, opinion leaders) as well as differentiate between opinion leaders and opinion leaders who have an additional capacity of opinion brokers.

Twitter represents of course a strategic digital space for the exercise of influence aiming at informing, offering opinions, building a community and creating personal [19] and due to "the possibility of asymmetric connection between participants, because users can follow other users without being followed (…) this imbalance alters the dynamic of influence"[19]. Traditional elites find new ways of cultivating their sphere of influence especially as expertise is key to justify the quality of opinion leader [21]. As an example, Figure 5-a, SpeakerPeloNetwork Graph allows us to identify the VP and White House as opinion leaders. This is not surprising given the checks-and-balances configuration characterizing the US governmental structure and their quality of institutional actors. Bravo and Del Valle [20] have established in their literature review concerning Politics and Twitter that traditional political elites understood as political institutions and political parties are sill leading the parliamentary networks online even though "it is not the same people that concentrate at the same time the highest number of followers, retweets, and mentions" (266). The literature shows that the most visible political leaders are also the most followed, mentioned and retweeted [20]; this is confirmed here as by virtue of her position as Speaker of the House, Pelosi enjoys high visibility.

Having said that, Twitter also opens opportunities for other less traditional actors to build a sphere of influence [20]. While personal characteristics are sometimes taken into account to gauge the influence of opinion leaders, their position in a network is often key to understanding their power and influence. This implies they do not necessarily need to be at the center but may act as opinion brokers to disseminate information across various groups [22]. Followers then in turn propagate these messages and are influenced by opinion leaders in their thoughts and attitudes; the most successful opinion leaders are those who see their messages disseminated by others [19]. On Twitter, the number of retweets can thus indicate the extent of an opinion leader's influence. There are thus two ways in which we can identify opinion leaders; in their study, Walter and Brugeman [22], when identifying opinion leaders at the COP21 conference, look for the centrality of their position in the network or the fact that they occupy brokering positions connecting different groups within a network. In link analysis, the density of the edges may point us to the brokering "quality" of the opinion leader. As an example, Figure 5-c highlights an opinion leader whose followers are not interactive with each other. It seems the prevalence of exclusive dyadic relations with the opinion leader over triadic relations (two followers and one opinion leader). Going back to the differentiation between opinion leaders and opinion leaders/brokers, this pattern may emphasize here an opinion leader strictly speaking. Figure 5-d also identifies three opinion leaders: however, ChefSkarapz and billcostmus are more influential opinion leaders than Brenwood Parrish in this network graph as the density of the graph suggests. We can speculate that ChefSkarapz and billcostmus also act as opinion brokers in addition to their centrality in the network.

IV. Conclusion

The aim of this research is to study radical views expressed on social media and how radical speech can be used for the radicalization of individuals online. In particular, the focus of the study is on extremism related to nationalism, hate speech, and supremacy.

A large dataset of more than 259,000 tweets posted over 5 years in relation to extreme nationalism was collected, and used in combination with machine learning and natural language processing techniques to gain insights about radicalization and extreme views. The results obtained are very promising and open the door for the ability to profile users based on their online activities and behavior, and provide early warning for the radicalization of individuals.

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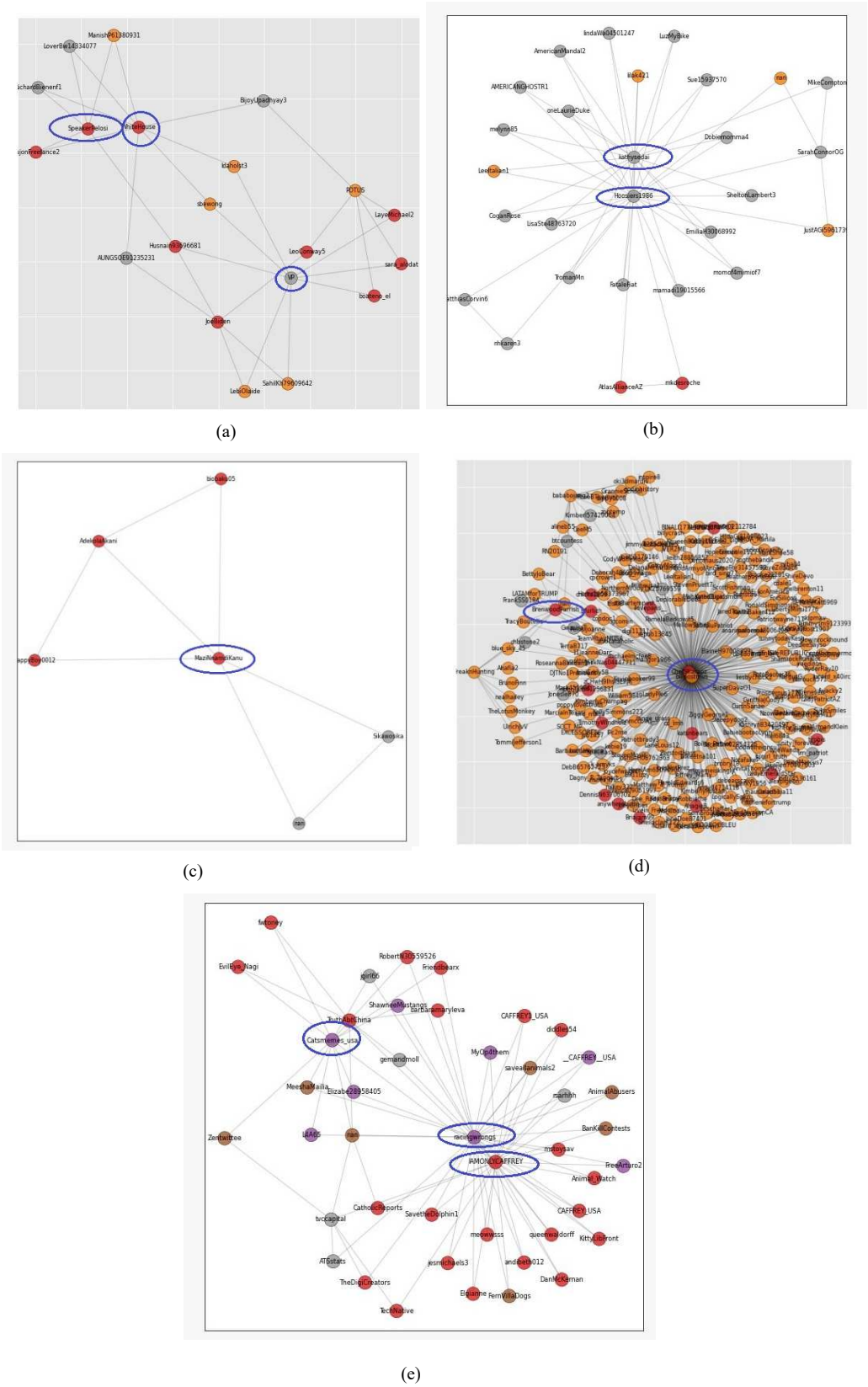

Fig. 5. Social circle analysis results for five opinion leaders: (a) SpeakerPelosi Network Graph; (b) Hoosiers1986 Network Graph; (c) MaziNnamdiKanu Network Graph; (d) BrenwoodParrish Network Graph; (e) IAMONLYCAFFREY Network Graph

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